Faculty of Computers and Information Technology

Department of Networks and Mobile

Graduation Project 1

❖ Course Information

Course Name: Graduation Project	**Course No:** ITNM4114
Prerequisite: -	**Credit Hours:**
Semester: Second 2025/2026	
Instructor: Hussam Musa. WhatsApp: 00972599291750, Email: hussam.s.musa@gmail.com	

❖ Description

This course allows students to apply the knowledge and skills acquired throughout the program to a real-world problem or research question within computer science networks and mobile. Students will be guided through all stages of the project lifecycle, from idea creation, strong project proposal, literature review to detailed project plan.

Students will undertake a substantial project, either individually or in small groups, focusing on a specific area of interest within the field, involving software engineering life cycle.

❖ Project Grade and Outline

Topic	Assessment Percentage	Expected Date
Chapter 1: Introduction	30%	2-May-2026
Chapter 2: Literature Review	30%	7-Jun-2026
Chapter 3: Methodology	40%	15-Jul-2026

❖ **More details will be discussed in lectures at these dates:**

Task	Date
Chapter 1: Introduction and Chapter 2: Literature Review	**5/Apr/2026 3pm**
Chapter 3: Methodology	**3/May/2026 3pm**

Table of Contents

Title Page

The title page is the first impression of your project. It must be clean, professional, and contain all required institutional information.

Required Elements

- **Project Title:** A clear, specific title (12–15 words max) that reflects the core focus. Example: "Design and Implementation of a Secure VPN Gateway Using OpenVPN and IPSec Protocols"
- **Student Name:** Full name as registered in university records
- **Student ID:** Official university registration number
- **Department:** Full department name (e.g., Department of Networks and Mobile)
- **University:** Complete university name and logo
- **Supervisor Name:** Academic title and full name
- **Submission Date:** Month and Year of final submission
- **Academic Year:** e.g., 2025 – 2026

✓ TIPS FOR A STRONG TITLE

Be specific — avoid vague titles like "Network Security Project"

Include the technology or method used (e.g., "Using Machine Learning", "Based on SDN")

Include the domain or application area (e.g., "for IoT Networks", "in Cloud Environments")

Good example: "Intrusion Detection System for IoT Networks Using Random Forest Classification"

Abstract

The abstract is a self-contained summary of your entire project in 150–250 words. A reader should understand your project fully from the abstract alone.

Structure Your Abstract in This Order

- **Context (1–2 sentences):** Set the scene: what area of CS/networking does this project belong to?
- **Problem (1–2 sentences):** What specific problem does your project address?
- **Objective (1 sentence):** What is the main goal?
- **Methodology (2–3 sentences):** What approach, tools, and techniques did you use?
- **Key Results (2–3 sentences):** What are the main findings? Include numbers where possible.
- **Conclusion (1 sentence):** What is the significance or impact?

✎ EXAMPLE ABSTRACT (Networking Project)

"Network security remains a critical challenge as organizations face increasingly sophisticated cyber threats. Current firewall systems often fail to detect zero-day attacks and advanced persistent threats (APTs) due to reliance on signature-based detection methods. This project aims to design and implement an AI-powered Intrusion Detection System (IDS) capable of identifying both known and unknown network attacks in real time. The system was developed using Python and Scikit-learn, trained on the CICIDS2017 dataset containing 2.8 million network flow records across 14 attack categories. A hybrid detection model combining Random Forest and LSTM neural networks was implemented and deployed on an Ubuntu server monitoring live network traffic via packet capture. The system achieved 97.3% detection accuracy with a false positive rate of only 1.8%, processing up to 12,000 packets per second. These results demonstrate that machine learning-based IDS can provide significantly better detection rates than traditional signature-based approaches while maintaining acceptable performance for medium-sized enterprise networks."

⚠ COMMON MISTAKES TO AVOID

Do NOT include references or citations in the abstract

Do NOT include figures or tables

Do NOT copy sentences from your introduction

Write the abstract LAST after completing all other chapters

Chapter 1: Introduction

The introduction sets the stage for your entire project. It should take the reader from a broad understanding of the field to the specific problem you are solving. Expected length: 5–8 pages.

1.1 Background and Context (1–2 pages)

Start broad, then narrow down to your specific area. Your goal is to convince the reader that this topic matters.

What to Write

- Introduce the general field (computer science, networking, cybersecurity, etc.)
- Discuss current trends, growth statistics, and real-world relevance
- Explain key technologies involved and why they matter today
- Gradually narrow the focus toward your specific problem area
- Include recent statistics or reports to show timeliness (e.g., "According to Cisco's 2024 Annual Report, global IP traffic will reach 4.8 ZB per year by 2025...")

✎ CS/NETWORKING EXAMPLE TOPICS

- Network security and intrusion detection in enterprise environments
- Software-Defined Networking (SDN) for campus or data center networks
- Cloud computing security and multi-tenant isolation
- IoT device management and protocol optimization
- Mobile application security and data protection
- Machine learning for network traffic classification
- Blockchain-based distributed systems for secure transactions
- 5G network slicing and quality of service management

1.2 Problem Statement (0.5–1 page)

This is the most critical part of your introduction. You must clearly define WHAT problem exists and WHY it needs solving.

How to Write a Strong Problem Statement

1. Identify the specific problem (not a general area, but a concrete issue)
2. Explain who is affected (network administrators, end users, organizations, etc.)
3. Describe current solutions and their shortcomings
4. State the gap your project fills
5. Quantify the problem if possible (e.g., "X% of networks are vulnerable to...")

✓ **EXAMPLE PROBLEM STATEMENT**

"Small and medium-sized enterprises (SMEs) often lack dedicated cybersecurity teams and rely on basic firewall configurations for network protection. According to the Verizon 2024 Data Breach Report, 43% of cyber-attacks target SMEs, yet only 14% are prepared to defend themselves. Existing commercial IDS solutions such as Snort and Suricata require significant expertise to configure and maintain, and their signature-based detection misses approximately 30% of novel attack patterns. This project addresses the need for an affordable, easy-to-deploy, AI-powered intrusion detection system that can automatically identify both known and zero-day attacks with minimal manual configuration."

1.3 Research Questions or Objectives (0.5 page)

State 3–5 specific, measurable objectives. Each must begin with an action verb and be achievable within your project scope.

Format Your Objectives Like This

6. To design a [system/architecture/protocol] that [specific capability]...
7. To implement [specific feature/algorithm] using [specific technology]...
8. To evaluate the performance of the proposed system in terms of [specific metrics]...
9. To compare the developed solution with existing approaches regarding [criteria]...

✎ **EXAMPLE OBJECTIVES (Network Security Project)**

1. To design a hybrid IDS architecture combining signature-based and anomaly-based detection
2. To implement the detection engine using Random Forest and LSTM algorithms in Python
3. To train and validate the model using the CICIDS2017 benchmark dataset
4. To evaluate detection accuracy, false positive rate, and processing throughput
5. To compare performance against Snort and Suricata on the same test dataset

1.4 Scope and Limitations (0.5 page)

Clearly define what IS and IS NOT included in your project. This manages expectations and shows maturity.

Scope (What IS Included)	Limitations (What IS NOT Included)
Features you will implement	Features deliberately excluded and why
Target platform/environment	Platforms not supported
Technologies and tools used	Technologies not covered

Scope (What IS Included)	Limitations (What IS NOT Included)
Types of testing performed	Edge cases not addressed
Dataset or traffic types analyzed	Data types not included
Network size/topology tested	Scale beyond your testing environment

1.5 Project Significance (0.5 page)

Explain why your project matters and who benefits from it.

- Academic contribution: What new knowledge or approach does it add?
- Practical value: Who can use this and how?
- Industry relevance: Does it solve a real-world problem?
- Advancement: How does it improve upon existing solutions?

1.6 Document Organization (0.5 page)

Provide a brief paragraph describing what each subsequent chapter contains to help readers navigate.

✓ EXAMPLE

"The remainder of this document is organized as follows: Chapter 2 reviews existing literature on intrusion detection systems and machine learning approaches. Chapter 3 describes the methodology including system design and testing framework. Chapter 4 details the implementation of the detection engine and web dashboard. Chapter 5 presents test results and performance analysis. Chapter 6 concludes with findings and recommendations for future work."

Chapter 2: Literature Review

The literature review demonstrates your understanding of existing knowledge and positions your work within the field. Expected length: 8–15 pages.

2.1 Introduction (0.5 page)

Explain the purpose and structure of your literature review. Mention databases searched (IEEE Xplore, ACM Digital Library, Google Scholar, ScienceDirect) and key search terms used.

2.2 Theoretical/Technical Foundation (2–3 pages)

Explain the fundamental concepts and technologies that underpin your project. Write for a reader who may not be an expert in your specific area.

What to Cover

- Core technical concepts (e.g., TCP/IP stack, encryption algorithms, routing protocols, virtualization)
- Key standards and protocols relevant to your project (e.g., IEEE 802.11, TLS 1.3, OSPF, BGP)
- Architectural patterns (e.g., client-server, microservices, SDN controller-switch model)
- Algorithms and data structures central to your work
- Mathematical or theoretical models if applicable

> ✎ **EXAMPLES BY PROJECT AREA**
>
> • Network Security: Explain cryptographic primitives, PKI, firewall types, IDS/IPS categories
>
> • SDN Project: Explain OpenFlow protocol, control plane vs data plane, SDN controllers
>
> • Cloud Computing: Explain virtualization, hypervisors, containers, orchestration
>
> • IoT: Explain MQTT, CoAP, edge computing, fog computing architectures
>
> • Machine Learning in Networking: Explain supervised vs unsupervised learning, feature extraction from network flows
>
> • Web Application: Explain MVC architecture, REST APIs, OWASP security principles

2.3 Review of Related Technologies and Tools (3–4 pages)

Discuss the technologies, frameworks, and tools relevant to your project. For each one:

- What is it and how does it work?
- What are its advantages and limitations?
- How does it compare with alternatives?
- Why is it relevant to your project?

Use Comparison Tables

Comparison tables are highly effective in CS/networking literature reviews.

Tool/Technology	Type	Strengths	Weaknesses	Relevance
Snort	Signature IDS	Mature, large rule set	Misses zero-day attacks	Baseline comparison
Suricata	Multi-threaded IDS	Fast, multi-threaded	High resource usage	Performance benchmark
Zeek (Bro)	Network analysis	Deep protocol analysis	Complex configuration	Feature extraction
Scikit-learn	ML framework	Easy to use, well-documented	Limited deep learning	Model training

2.4 Review of Related Work (3–4 pages)

This is the most important section of the literature review. Discuss 10–20 relevant research papers and projects.

For Each Study, Include

10. Full citation (Author, Year, Title)
11. Brief description of what they did
12. Methodology and tools used
13. Key results and findings
14. Strengths of their approach
15. Limitations or gaps
16. How your project differs or improves upon it

✓ HOW TO CRITICALLY ANALYZE (Not Just Summarize)

Do NOT just describe each paper — analyze and compare them

Group related studies together by theme or approach

Identify patterns: What methods are most popular? What metrics are used?

Find contradictions: Do different studies report conflicting results?

Spot trends: Is the field moving toward a particular approach?

2.5 Gap Analysis and Justification (1 page)

Synthesize all limitations identified above into a clear statement of what is missing and how your project fills this gap.

Structure

17. Summarize common limitations across reviewed studies
18. Identify the specific gap your project addresses
19. Explain why this gap matters
20. State how your project fills it
21. Create a summary comparison table showing your project vs. existing work

2.6 Summary (0.5 page)

Recap the main themes from the literature review, reinforce the gap, and transition naturally into the methodology chapter.

> ⚠ **REFERENCE REQUIREMENTS**
>
> Minimum 20–30 references for undergraduate projects
>
> At least 80% should be from the last 5 years (2020–2025)
>
> Prefer peer-reviewed journals and conference papers (IEEE, ACM, Springer, Elsevier)
>
> Avoid Wikipedia, blog posts, and unverified online sources as primary references
>
> Use a consistent citation style (IEEE recommended for CS/networking)

Chapter 3: Methodology

This chapter explains HOW you designed, built, and tested your project. A reader should be able to replicate your work from this description. Expected length: 10–15 pages.

3.1 Research Design and Approach (1–2 pages)

Explain the overall approach and development methodology.

- Type of project: Developmental (building a system), Experimental (testing a hypothesis), Comparative (evaluating approaches), or Hybrid
- Development model: Agile, Waterfall, Iterative, Prototyping — justify your choice
- Project phases: Outline the major stages of your work

3.2 Requirements Analysis (2–3 pages)

3.2.1 Functional Requirements

List what the system MUST do. Use a structured format:

Req. ID	Requirement Description	Priority
FR-01	The system shall capture and analyze network packets in real time	High
FR-02	The system shall classify traffic as normal or attack with >95% accuracy	High
FR-03	The system shall generate alerts with attack type, source IP, timestamp	High
FR-04	The system shall provide a web-based dashboard for monitoring	Medium
FR-05	The system shall log all detected events to a database	Medium
FR-06	The system shall support rule updates without system restart	Low

3.2.2 Non-Functional Requirements

Req. ID	Category	Requirement	Target Value
NFR-01	Performance	Packet processing throughput	≥10,000 packets/sec
NFR-02	Performance	Alert generation latency	<2 seconds
NFR-03	Reliability	System uptime	99.5%
NFR-04	Security	Dashboard authentication	Multi-factor auth
NFR-05	Scalability	Concurrent monitored hosts	≥100 hosts
NFR-06	Usability	Dashboard learning time	<30 minutes

3.3 System Design (3–4 pages)

3.3.1 System Architecture

Provide a high-level architecture diagram and explain each component:

- What architectural pattern are you using? (client-server, layered, microservices, SDN, etc.)
- Draw a clear diagram showing all components and their interactions
- Explain the role of each component
- Describe communication protocols between components (HTTP/REST, gRPC, MQTT, raw sockets, etc.)
- Explain data flow through the system from input to output

> ✎ **ARCHITECTURE EXAMPLES FOR CS/NETWORKING**
>
> • IDS: Packet Capture → Feature Extraction → ML Engine → Alert Manager → Dashboard
>
> • SDN: Applications → SDN Controller (e.g., ONOS/Floodlight) → OpenFlow Switches → Network Devices
>
> • Web App: Browser → React Frontend → REST API (Spring Boot/Flask) → Database (MySQL/MongoDB)
>
> • IoT: Sensors → MQTT Broker → Edge Gateway → Cloud Processing → Dashboard

3.3.2 Database Design

Include ER diagrams, table schemas, field types, constraints, and relationships. Explain your database choice (relational vs NoSQL) and why it suits your project.

3.3.3 Network Design (for Networking Projects)

If your project involves network infrastructure:

- Network topology diagram (physical and logical)
- IP addressing scheme and subnet plan
- VLAN configuration and segmentation strategy
- Routing protocol selection and configuration
- Security zones and firewall placement
- Simulation environment setup (GNS3, Cisco Packet Tracer, Mininet, etc.)

3.3.4 Interface Design

Include wireframes or mockups of your user interface. Explain navigation flow and design decisions.

3.4 Technology Stack (1–2 pages)

Document every technology used with justification:

Category	Technology	Version	Purpose / Justification
Programming Language	Python	3.11	Rich ML libraries, networking support
ML Framework	Scikit-learn / TensorFlow	1.3 / 2.15	Model training and evaluation
Packet Capture	Scapy / Pyshark	2.5	Network traffic capture and parsing
Web Framework	Flask / Django	3.0	Dashboard backend API
Frontend	React.js	18.2	Interactive monitoring dashboard
Database	PostgreSQL / MongoDB	16 / 7.0	Event storage and querying
Simulation	Mininet / GNS3	2.3	Network topology emulation
Version Control	Git + GitHub	--	Source code management
IDE	VS Code / PyCharm	--	Development environment
OS	Ubuntu 22.04 LTS	--	Development and deployment

3.5 Data Collection Methods (1–2 pages)

Explain what data you need and how you will collect it:

- For ML projects: Dataset source (CICIDS2017, NSL-KDD, custom capture), size, features, and preprocessing steps
- For performance testing: What metrics to measure, how to collect them (automated scripts, Wireshark, iperf, etc.)
- For user studies: Survey instruments, participant recruitment, data collection procedures

3.6 Testing Strategy (1–2 pages)

Define your complete testing plan including unit tests, integration tests, system tests, performance tests, and security tests. Specify success criteria for each test type.

3.7 Project Timeline

Include a Gantt chart or timeline table showing project phases, milestones, and deliverables.

By following this structure, you'll ensure your proposal is well-organized and covers all the important aspects that likely to be expected.